

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (EC)

April- 2013

EC 308 Antenna & Wave Propagation

Date: 30.04.2013, Tuesday

Time: 10:00 p.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1** (a) Calculate the maximum effective aperture of a microwave antenna which has directivity of 900. 1
- (b) A thin dipole antenna is $\lambda/15$ long, if its loss resistance is 1.5Ω . Find R_r and efficiency 2
- (c) The radiation resistance of an antenna is 67Ω , loss resistance 3Ω . If antenna has power gain 16 dB then find directivity. 1
- (d) Explain in brief multipath fading? 1
- (e) What is the skip distance of a radio wave? 1
- (f) When a wide band of frequencies is transmitted simultaneously, each frequency will vary in the amount of fading. What is this variable fading called? 1
- Q - 2** (a) Explain principal of pattern multiplication. Using the pattern multiplication principle, explain the radiation pattern of 4 isotropic elements fed in phase, spaced $\lambda/2$ 7
- (b) Design a broadside Dolph-Tschebyscheff array of 4 elements with spacing $\lambda/2$ between the elements and with a major to minor lobe ratio of 19.1 dB. Find the excitation coefficients and form the Array factor. 7

OR

- Q - 2** (a) Starting from retarded currents derive an expression for E and H components for a short dipole antenna if the spherical system is defined in r, θ and Φ . 10
- (b) Sketch the helical geometry with its associated dimension showing relationship between circumference, spacing, turn length and pitch angle of helix 4
- Q - 3** (a) In far field, for a small Loop Antenna derive field equations. Also derive Radiated Power and Radiation Resistance equation. 10
- (b) Determine the distance from elementary dipole operating at 1 MHz at which radiation field is 6 times the induction field 4

OR

- Q - 3** (a) Explain different layers of ionosphere 5
- (b) Prove that $R_r = 73 \Omega$ in $\lambda/2$ antenna 5
- (c) Derive equation for Friis transmission formula. 4

SECTION - II

- Q - 4** (a) What is an isotropic radiator? 1
- (a) An antenna that radiates equally in all directions (b) An antenna with equal dimensions (c) An antenna designed to broadcast in the troposphere (d) None of the above

	(b) Which type of antenna would typically have the highest directivity? (a) 1/4-wave whip (b) Folded dipole (c) Yagi (d) Parabolic	1
	(c) Yagi antennas are constructed of which three types of elements? (a) Transmitter, receiver, and transceiver (b) Underdriven, driven, and overdriven (c) Reflector, driven, and director (d) Aluminum, iron, and magnesium	1
	(d) If antenna directivity and antenna gain are equal, then antenna efficiency is _____. (a) 20 (b) 50 (c) 75 (d) 100	1
	(e) For a typical microstrip antenna, the height of the substrate is usually: (a) much smaller than the width of the patch (b) about the same as the width of the patch (c) greater than the width of the patch	1
	(f) The input impedance of a rectangular microstrip fed at the center of its length is, ideally: (a) Open circuit (b) short circuit (c) an impedance of finite value	1
	(g) For a paraboloidal reflector, the amplitude distribution over an aperture plane perpendicular to the axis of the reflector _____ as the observation point moves from the center towards the edges of the aperture plane. (a) Decreases (b) increases (c) stays same	1
Q - 5	(a) Design a rectangular patch antenna at frequency 1 GHz using substrate with dielectric constant of 2.2, height of substrate $h=0.1588\text{cm}$.	8
	(b) Design 3 element Yagi Uda antenna at 20 GHz	3
	(c) Design Resonant antenna (leg length, tilt angle, height above ground) at frequency 22 GHz and β is 35° .	3
OR		
Q - 5	(a) Explain how log periodic antenna works as broadband antenna.	5
	(b) Explain Babinet's Principle. And discuss it for complementary antenna	4
	(c) Design pyramidal horn antenna (length, H plane aperture and flare angles) at 1GHz. The given parameters are $a_E=10\lambda$, $\delta=0.2\lambda$, in E plane and 0.375λ in H plane. Also calculate directivity.	5
Q - 6	(a) Explain measurement of radiation pattern antenna under test	5
	(b) Explain in brief Polyrod	2
	(c) Describe various types of horn used at microwave frequency	7
OR		
Q - 6	(a) Estimate the diameter and effective aperture of a parabolic reflector antenna required to produce null beam width of 15 degree at 5 GHz.	5
	(b) Differentiate cassegrain feed and offset feed system in reflector antenna	4
	(c) Explain the SWR method for the impedance measurement of antenna	5